

LESSON 4 : PERCENTAGES II (A)

Checkpoint 1

The amount of rainfall of a city was 2500 mm in 2019. The amount of rainfall in 2020 was increased by 12%, and in 2021, it was decreased by 15%. Find the amount of rainfall in 2021.

∴ The amount of rainfall in 2021,
 $2500 (1 + 12\%) (1 - 15\%)$
 $= 2094.4 \text{ mm}$

Rainfall (mm)	
2019	2500
2020	2500 (1 + 12%)
2021	2500 (1 + 12%) (1 - 15%)

1. Find the new value in each of the following.

- (a) 200 increases by 30% and then increases by 40%.
- (b) 8 m decreases by 50% and then decreases by 10%.
- (c) \$1 600 decreases by 25% and then increases by 60%.

- (a) New value = $200 \times (1 + 30\%) \times (1 + 40\%)$
 $= 364$
- (b) New value = $8 \times (1 - 50\%) \times (1 - 10\%) \text{ m}$
 $= 3.6 \text{ m}$
- (c) New value = $\$1\,600 \times (1 - 25\%) \times (1 + 60\%)$
 $= \$1\,920$

2. The amount of rainfall of a city was 2 200 mm in 2010. The amount of rainfall in 2011 was decreased by 18%, and in 2012, it was increased by 2%. Find the amount of rainfall in 2012.

$$\begin{aligned}
 \text{Amount of rainfall in 2012} &= 2200 \times (1 - 18\%) \times (1 + 2\%) \text{ mm} \\
 &= 2200 \times 0.82 \times 1.02 \text{ mm} \\
 &= 1840.08 \text{ mm}
 \end{aligned}$$

3. There are 200 books on a bookshelf, of which 60% are fiction and the rest are non-fiction. If the numbers of fiction and non-fiction books are increased by 10% and 20% respectively, find

- (a) the number of fiction books now,
- (b) the number of non-fiction books now,
- (c) the increase in the number of books on the bookshelf.

$$\begin{aligned}
 \text{(a) Number of fiction books now} &= 200 \times 60\% \times (1 + 10\%) \\
 &= 200 \times 0.6 \times 1.1 \\
 &= 132
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) Number of non-fiction books now} &= 200 \times (1 - 60\%) \times (1 + 20\%) \\
 &= 200 \times 0.4 \times 1.2 \\
 &= 96
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) The increase in the number of books on the bookshelf} &= (132 + 96) - 200 \\
 &= 28
 \end{aligned}$$

Checkpoint 2

Tony's monthly salary was increased by 4% and 3.5% in last two years respectively. If Tony's monthly salary is \$13 455 in 2020, find his monthly salary 2 years ago (i.e. in 2018).

Salary (\$)	
2018	x
2019	$x(1 + 4\%)$
2020	$x(1 + 4\%)(1 + 3.5\%) = \$13\,455$

4. A table is worth \$697 now. Its current value is 18% lower than that of last year. Last year, its value was 15% lower than that of 2 years ago. Find the value of the table 2 years ago.

Let \$ n be the value of the table 2 years ago.

$$n \times (1 - 15\%) \times (1 - 18\%) = 697$$

$$n = 697 \div 0.85 \div 0.82$$

$$= 1000$$

$\therefore$ 2 years ago, the value of the table was \$1000.

5. When a number is first increased by 15% and then decreased by 30%, the result is 241.5. What is the original value of the number?

$$y \times (1 + 15\%) \times (1 - 30\%) = 241.5$$

$$y = 300$$

6. If a number is increased by 25% and then decreased by 10%, it becomes 99. Find the original number.

$$y \times (1 + 25\%) \times (1 - 10\%) = 99$$

$$y = 88$$

7. The monthly salary of Mr. Lee was increased by 12% and 25% in the last two years. If his current monthly salary is \$17 500, what was his monthly salary two years ago?

$$x \times (1 + 12\%) \times (1 + 25\%) = 17\,500$$

$$x = 12\,500$$

8. Find the original value in each of the following.

(a) After increasing by 6% and then increasing by 7%, the new value is 90 736.

(b) After increasing by 30% and then decreasing by 20%, the new value is 208 g.

(c) After decreasing by 40% and then increasing by 60%, the new value is 76.8 cm³.

(a) Let P be the original value.

$$90\,736 = P(1 + 6\%)(1 + 7\%)$$

$$90\,736 = P(1.06)(1.07)$$

$$P = \frac{90\,736}{(1.06)(1.07)}$$

$$= 80\,000$$

$\therefore$ The original value is 80 000.

(b) Let P g be the original value.

$$208 = P(1 + 30\%)(1 - 20\%)$$

$$208 = P(1.3)(0.8)$$

$$P = \frac{208}{(1.3)(0.8)}$$

$$= 200$$

$\therefore$ The original value is 200 g.

(c) Let P cm³ be the original value.

$$76.8 = P(1 - 40\%)(1 + 60\%)$$

$$76.8 = P(0.6)(1.6)$$

$$P = \frac{76.8}{(0.6)(1.6)}$$

$$= 80$$

$\therefore$ The original value is 80 cm³.

9. The price of a car is \$125 000. The price is first decreased by 15% and then further decreased by 8%. Find the overall percentage change in the price of the car.

$$\begin{aligned}\text{New price of the car} &= \$125\,000 \times (1 - 15\%) \times (1 - 8\%) \\ &= \$125\,000 \times 0.85 \times 0.92 \\ &= \$97\,750\end{aligned}$$

$$\begin{aligned}\text{Overall percentage change} &= \frac{97\,750 - 125\,000}{125\,000} \times 100\% \\ &= \underline{\underline{-21.8\%}}\end{aligned}$$

Recall:

$$\text{Percentage change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

10. The volume of water in the container is 1.25L. The volume is first increased by 22% and then further decreased by 30%. Find the overall percentage change in the volume of water.

$$\begin{aligned}\text{New volume of water} &= 1.25 \times (1 + 22\%) \times (1 - 30\%) \text{ L} \\ &= 1.25 \times 1.22 \times 0.7 \text{ L} \\ &= 1.0675 \text{ L} \\ \text{Overall percentage change} &= \frac{1.0675 - 1.25}{1.25} \times 100\% \\ &= \underline{\underline{-14.6\%}}\end{aligned}$$

11. The tax revenue of a government increased by 20%, 10%, 35% respectively in the first three quarters of 2021, and then decreased by 45% in the fourth quarter of the year. What was the overall percentage change in the quarterly tax revenue over the whole year 2021?

Let \$P\$ be the tax revenue of the government in the fourth quarter of 2020.

$$\begin{aligned}\text{Tax revenue in the fourth quarter of 2021} &= \$P(1 + 20\%)(1 + 10\%)(1 + 35\%)(1 - 45\%) \\ &= \$P(1.2)(1.1)(1.35)(0.55) \\ &= \$0.9801P\end{aligned}$$

$$\begin{aligned}\text{Percentage change} &= \frac{0.9801P - P}{P} \times 100\% \\ &= \frac{-0.0199P}{P} \times 100\% \\ &= \underline{\underline{-1.99\%}}\end{aligned}$$

12. If a number is decreased by 90% and then increased by $x\%$, the overall percentage change is -83% . Find x .

$$\begin{aligned}\text{Let } P \text{ be the original number.} \\ P(1 - 90\%)(1 + x\%) &= P(1 - 83\%) \\ P(0.1)\left(1 + \frac{x}{100}\right) &= 0.17P \\ 1 + \frac{x}{100} &= 1.7 \\ \frac{x}{100} &= 0.7 \\ x &= \underline{\underline{70}}\end{aligned}$$

13. The length, the width and the height of a rectangular block are 15 cm, 8 cm, and 18 cm respectively. If the length increases by 20% and the width increases by 15%, but the volume decreases by 10%, find the percentage change in height.

Let $n\%$ be the percentage change in height.

$$15(1 + 20\%) \times 8(1 + 15\%) \times 18(1 + n\%) = 15 \times 8 \times 18 \times (1 - 10\%), n\% = \frac{0.9}{(1.2)(1.15)} - 1, \\ n = -34.8 \quad \text{Ans. The percentage decrease in height is 34.8\%}$$

14. The length, the width and the height of a rectangular block are 10 cm, 15 cm, and 21 cm respectively. If the length decreases by 15% and the width increases by 16%, but the volume increases by 6%, find the percentage change in height.

Let $n\%$ be the percentage change in height,

$$\frac{[10(1-15\%) \cdot 15(1+16\%) \cdot 21(1+n\%)] - (10 \cdot 15 \cdot 21)}{10 \cdot 15 \cdot 21} = 6\%$$

$$8.5(17.4)[21(1+n\%)] = 3339$$

$$1+n\% = 1.0751$$

$$n\% = +7.51\%$$

$$\therefore \text{The percentage change of height is } 7.51\%$$

15. The radius r cm of a circle is decreased by 20%.
- Express the original area of the circle in terms of r .
 - Express the new area of the circle in terms of r .
 - Find the percentage change in the area of the circle.

(a) Original area of the circle = $\pi r^2 \text{ cm}^2$

(b) New radius of the circle = $r \times (1 - 20\%) \text{ cm} = 0.8r \text{ cm}$
 New area of the circle = $\pi(0.8r)^2 \text{ cm}^2$

$$= 0.64\pi r^2 \text{ cm}^2$$

(c) Percentage change in the area of the circle

$$= \frac{0.64\pi r^2 - \pi r^2}{\pi r^2} \times 100\%$$

$$= -36\%$$

Hints:

$$\frac{3x - x}{x} = \frac{3x^2 - x^2}{x^2}$$

$$= \frac{2x}{x} = \frac{2x^2}{x^2}$$

$$= 2 = 2$$

16. The length of a square is l cm. If the length is first increased by 20% and then further increased by 20%,
- express the new area of the square in terms of l ,
 - find the percentage change in the area of the square.

(a) New area of the square

$$= [l \times (1 + 20\%) \times (1 + 20\%)]^2 \text{ cm}^2$$

$$= (l \times 1.2 \times 1.2)^2 \text{ cm}^2$$

$$= 2.0736l^2 \text{ cm}^2$$

(b) Percentage change in the area of the square

$$= \frac{2.0736l^2 - l^2}{l^2} \times 100\%$$

$$= +107.36\%$$

17. Evelyn's pocket money was increased by 10% last month, but is decreased by 10% this month. What is the overall percentage change of her pocket money?

[Hint: Let \$x be Evelyn's original pocket money.]

$$\begin{aligned}
 &\text{Let } \$x \text{ be Evelyn's original pocket money.} \\
 &\text{New amount of Evelyn's pocket money} \\
 &= \$x \times (1 + 10\%) \times (1 - 10\%) \\
 &= \$x \times 1.1 \times 0.9 \\
 &= \$0.99x \\
 &\text{Overall percentage change of her pocket money} \\
 &= \frac{0.99x - x}{x} \times 100\% \\
 &= \underline{\underline{-1\%}}
 \end{aligned}$$

18. The length and the width of a rectangle are l cm and w cm respectively. If the length is increased by 10% and the width is decreased by 30%,

- (a) express the new area of the rectangle in terms of l and w ,
(b) find the percentage change in the area of the rectangle.

$$\begin{aligned}
 \text{(a) New area of the rectangle} \\
 &= l \times (1 + 10\%) \times w \times (1 - 30\%) \text{ cm}^2 \\
 &= 1.1l \times 0.7w \text{ cm}^2 \\
 &= \underline{\underline{0.77lw \text{ cm}^2}} \\
 \text{(b) Original area of the rectangle} &= lw \text{ cm}^2 \\
 \text{Percentage change in the area of the rectangle} \\
 &= \frac{0.77lw - lw}{lw} \times 100\% \\
 &= \underline{\underline{-23\%}}
 \end{aligned}$$

19. The angle of a sector is decreased by 20% but its radius is increased by $k\%$. If the arc length of the sector remains unchanged, find the value of k .

$$\begin{aligned}
 &\text{Let } \theta \text{ and } r \text{ be the original angle and the original radius of the sector respectively.} \\
 &\text{Original arc length} = \frac{\theta}{360^\circ} \times 2\pi r \\
 &\text{New angle} = \theta \times (1 - 20\%) \\
 &\quad = 0.8\theta \\
 &\text{New radius} = r \times (1 + k\%) \\
 &\text{New arc length} = \frac{0.8\theta}{360^\circ} \times 2\pi \times r \times (1 + k\%) \\
 &\therefore \text{Arc length of the sector remains unchanged.} \\
 &\therefore \frac{\theta}{360^\circ} \times 2\pi r = \frac{0.8\theta}{360^\circ} \times 2\pi \times r \times (1 + k\%) \\
 &\quad 1.25 = 1 + k\% \\
 &\quad 0.25 = k\% \\
 &\quad k = \underline{\underline{25}}
 \end{aligned}$$

20. In a normal day, Helen spends 37.5% of the time sleeping and 10% of the remaining time studying.
- What is the amount of time used for other purposes in a normal day?
 - An examination is approaching, so Helen has to spend more time on studying. She has decided to reduce the sleeping time and other purposes by 5% respectively.
 - Find the percentage change in Helen's studying time.
 - Furthermore, she increases the reading speed by 4%. Find the percentage increase in the amount of reading materials she can read.

$$\begin{aligned} \text{(a) Amount of time used for other purposes} &= 24 \times (1 - 37.5\%) \times (1 - 10\%) \\ &= 13.5 \text{ hours} \end{aligned}$$

$$\begin{aligned} \text{(bi) Original amount of studying time} &= 24 \times (1 - 37.5\%) \times 10\% \\ &= 1.5 \text{ hours} \\ \text{New amount of studying time} &= 24 - 24 \times 37.5\% \times (1 - 5\%) - 13.5 \times (1 - 5\%) \\ &= 2.625 \text{ hours} \\ \text{\% change} &= \frac{2.625 - 1.5}{1.5} \times 100\% \\ &= 75\% \end{aligned}$$

$$\begin{aligned} \text{(ii) Let } x \text{ hour be the original rate of reading.} \\ \text{\% increase} &= \frac{2.625 \times (1 + 4\%)x - 1.5x}{1.5x} \times 100\% \\ &= 82\% \end{aligned}$$

21. The operating cost of a company is composed of three parts:

Advertising \$10 000 Rent \$40 000 Wage \$200 000

If the advertising cost increases by 10%, the rent decreases by 20% and the wage increases by 5%, find the percentage change in the operating cost of the company.

$$\begin{aligned} \text{Original operating cost} &= \$ (10\,000 + 40\,000 + 200\,000) \\ &= \$250\,000 \\ \text{New advertising cost} &= \$10\,000 \times (1 + 10\%) \\ &= \$11\,000 \\ \text{New rent} &= \$40\,000 \times (1 - 20\%) \\ &= \$32\,000 \\ \text{New wage} &= \$200\,000 \times (1 + 5\%) \\ &= \$210\,000 \end{aligned}$$

$$\begin{aligned} \text{New operating cost} &= \$ (11\,000 + 32\,000 + 210\,000) \\ &= \$253\,000 \end{aligned}$$

$$\begin{aligned} \text{Percentage change in the operating cost} &= \frac{253\,000 - 250\,000}{250\,000} \times 100\% \\ &= \underline{+1.2\%} \end{aligned}$$

22. Last week, the number of books sold in bookstores P , Q and R were 150, 240 and 320 respectively. As compared to last week, the number of books sold this week in bookstore P decreases by 2%, that in bookstore Q increases by 5% and that in bookstore R remains unchanged. Find the percentage change in the total number of books sold in the three bookstores, correct to 3 significant figures.

$$\begin{aligned}\text{Number of books sold in the three bookstores last week} \\ &= 150 + 240 + 320 \\ &= 710\end{aligned}$$

$$\begin{aligned}\text{number of books sold in bookstore } P \\ &= 150 \times (1 - 2\%) \\ &= 147 \\ \text{number of books sold in bookstore } Q \\ &= 240 \times (1 + 5\%) \\ &= 252\end{aligned}$$

$$\begin{aligned}\text{total number of books sold in the three bookstores} \\ &= 147 + 252 + 320 \\ &= 719\end{aligned}$$

$$\begin{aligned}\therefore \text{Percentage change} &= \frac{719 - 710}{710} \times 100\% \\ &= \underline{+1.27\%, \text{ cor. to 3 sig. fig.}}\end{aligned}$$

23. Last week, the numbers of male customers and the female customers in a shopping mall were in the ratio 1 : 3. This week, the number of male customers increases by 16% while the number of female customers decreases by 12%. Is there an increase in the total number of customers as compared to last week? Explain your answer.

$$\begin{aligned}\text{Let } x \text{ be the total number of customers last week.} \\ \text{Last week,} \\ \text{number of male customers} &= \frac{1}{1+3} x \\ &= 0.25x \\ \text{number of female customers} &= \frac{3}{1+3} x \\ &= 0.75x \\ \text{This week,} \\ \text{number of male customers} &= 0.25x(1 + 16\%) \\ &= 0.29x \\ \text{number of female customers} &= 0.75x(1 - 12\%) \\ &= 0.66x \\ \text{total number of customers} &= 0.29x + 0.66x = 0.95x\end{aligned}$$

$$\begin{aligned}\therefore \text{Percentage change} &= \frac{0.95x - x}{x} \times 100\% \\ &= \frac{-0.05x}{x} \times 100\% \\ &= -5\% \\ \therefore \text{The total number of customers does not} \\ &\quad \underline{\text{increase as compared to last week.}}\end{aligned}$$

24. The numbers of tourists of three cities A , B and C in 2020 are listed below:

City	A	B	C
Number of tourists	2 500	900	1 600

In 2021, the numbers of tourists of cities B and C increased by 40% and 20% respectively. If the total number of tourists of the three cities increased by 4%, find the percentage change in the number of tourists of city A .

$$\begin{aligned}\text{Total number of tourists of the three cities} \\ &= 2\,500 + 900 + 1\,600 \\ &= 5\,000 \\ \text{Let } x\% \text{ be the percentage change in the number of} \\ &\text{tourists of city } A. \\ \text{In 2021,} \\ \text{number of tourists of city } A &= 2\,500 \times (1 + x\%) \\ \text{number of tourists of city } B &= 900 \times (1 + 40\%) \\ &= 1\,260 \\ \text{number of tourists of city } C &= 1\,600 \times (1 + 20\%) \\ &= 1\,920\end{aligned}$$

$$\begin{aligned}\text{total number of tourists in the three cities} \\ &= 5\,000 \times (1 + 4\%) \\ &= 5\,200 \\ \therefore 2\,500 \times (1 + x\%) + 1\,260 + 1\,920 &= 5\,200 \\ 2\,500 \times (1 + x\%) &= 2\,020 \\ 1 + \frac{x}{100} &= \frac{2\,020}{2\,500} \\ \frac{x}{100} &= -0.192 \\ x &= -19.2 \\ \therefore \text{The percentage change in the number of} \\ &\quad \underline{\text{tourists of city } A \text{ was } -19.2\%}.\end{aligned}$$

25. There are some apples, oranges and peaches in a box. The numbers of apples, oranges and peaches are in the ratio 2 : 5 : 3. If the number of apples decreases by 40%, the number of oranges increases by 15% and the number of peaches decreases by 5%, find the percentage change in the number of fruits in the box.

Let x be the total number of fruits in the box.

Original number of apples

$$= \frac{2}{2+5+3}x$$

$$= 0.2x$$

Original number of oranges

$$= \frac{5}{2+5+3}x$$

$$= 0.5x$$

Original number of peaches

$$= \frac{3}{2+5+3}x$$

$$= 0.3x$$

New number of apples

$$= 0.2x \times (1 - 40\%)$$

$$= 0.12x$$

New number of oranges

$$= 0.5x \times (1 + 15\%)$$

$$= 0.575x$$

New number of peaches

$$= 0.3x \times (1 - 5\%)$$

$$= 0.285x$$

New number of fruits

$$= 0.12x + 0.575x + 0.285x$$

$$= 0.98x$$

$$\therefore \text{Percentage change} = \frac{0.98x - x}{x} \times 100\%$$

$$= \frac{-0.02x}{x} \times 100\%$$

$$= \underline{\underline{-2\%}}$$

26. If the height of a right cylinder increases by 60% and its curved surface area remains unchanged, find the percentage change in the volume of the cylinder.

Let r , h and R be the original base radius, original height and new base radius of the cylinder respectively.

$$\text{New height of the cylinder} = h(1 + 60\%)$$

$$= 1.6h$$

$\therefore$ Original curved surface area = new curved surface area

$$\therefore 2\pi rh = 2\pi R(1.6h)$$

$$R = 0.625r$$

Original volume of the cylinder = $\pi r^2 h$

$$\text{New volume of the cylinder} = \pi R^2(1.6h)$$

$$= \pi(0.625r)^2 \times (1.6h)$$

$$= 0.625\pi r^2 h$$

$$\therefore \text{Percentage change in the volume of the cylinder} = \frac{0.625\pi r^2 h - \pi r^2 h}{\pi r^2 h} \times 100\%$$

$$= -\frac{0.375\pi r^2 h}{\pi r^2 h} \times 100\%$$

$$= \underline{\underline{-37.5\%}}$$